

ComicCraftAI

AI-Powered Comic Story Creator

Using Gemini Models

Project Documentation

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1. Project Overview

ComicCraftAI is an AI-powered web application designed to transform a simple story idea into a structured comic.

The user provides a story concept along with details such as the **main character, setting, tone, and art style**. ComicCraftAI then develops the idea into a **five-panel comic**, generates the story content and illustrations, presents the completed comic through a web interface, and allows the result to be exported as a PDF.

The main idea behind the project is to make comic creation easier for people who may have a creative idea but do not have advanced skills in **story writing, illustration, or comic design**.

Our implementation combines AI-based text generation with image generation and a Python-based web application to create an end-to-end creative workflow.

2. Problem / Need for the Project

Creating a comic traditionally involves several separate tasks:

- Developing a story
- Dividing the story into scenes
- Writing narration and dialogue
- Designing characters and scenes
- Creating illustrations
- Arranging the panels

- Preparing the final comic for sharing or printing

For beginners, managing all these stages can be time-consuming and technically difficult.

ComicCraftAI addresses this by bringing these stages together into one workflow. Instead of starting with a blank page and creating everything manually, the user can begin with a simple idea and allow the application to assist with the creative development.

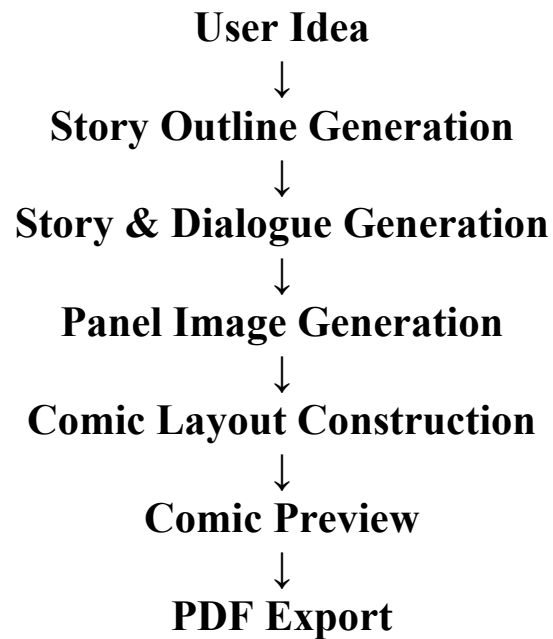
3. Objectives

The main objectives of ComicCraftAI are:

1. To convert a simple user idea into a structured comic story.
2. To organize the story into five connected panels.
3. To generate narration and dialogue suitable for each panel.
4. To create AI-generated illustrations based on the individual scenes.
5. To provide customization through character, setting, tone, and art style.
6. To display the generated comic through an easy-to-use web interface.
7. To combine the generated content into a downloadable PDF.
8. To demonstrate how different AI services can be integrated into a single application

4. How ComicCraftAI Works

The application follows this overall workflow:



The system separates the creative process into stages instead of asking one AI model to perform everything at once.

5. Key Features

- ◆ *AI Story Generation*
- ◆ *Five-Panel Story Structure*
- ◆ *Customizable Story Inputs*
- ◆ *AI-Generated Illustrations*
- ◆ *Integrated Comic Preview*
- ◆ *PDF Export*

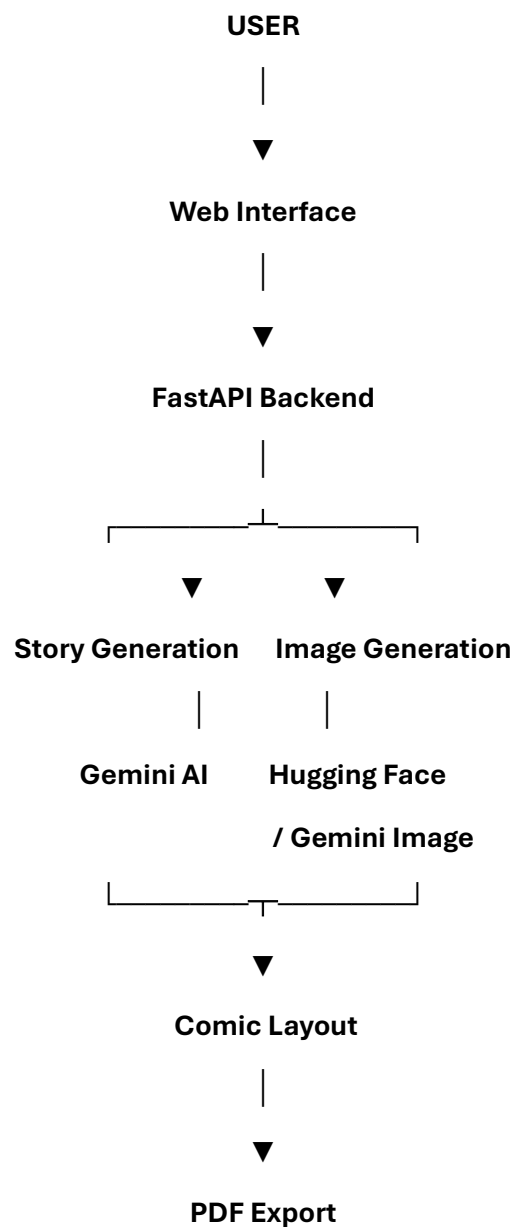
6. Technology Stack

Python , FastAPI , Jinja2 , Google Gemini , Hugging Face Interface , Pillow , FPDF2 , HTML & CSS , Git & GitHub , VS Code

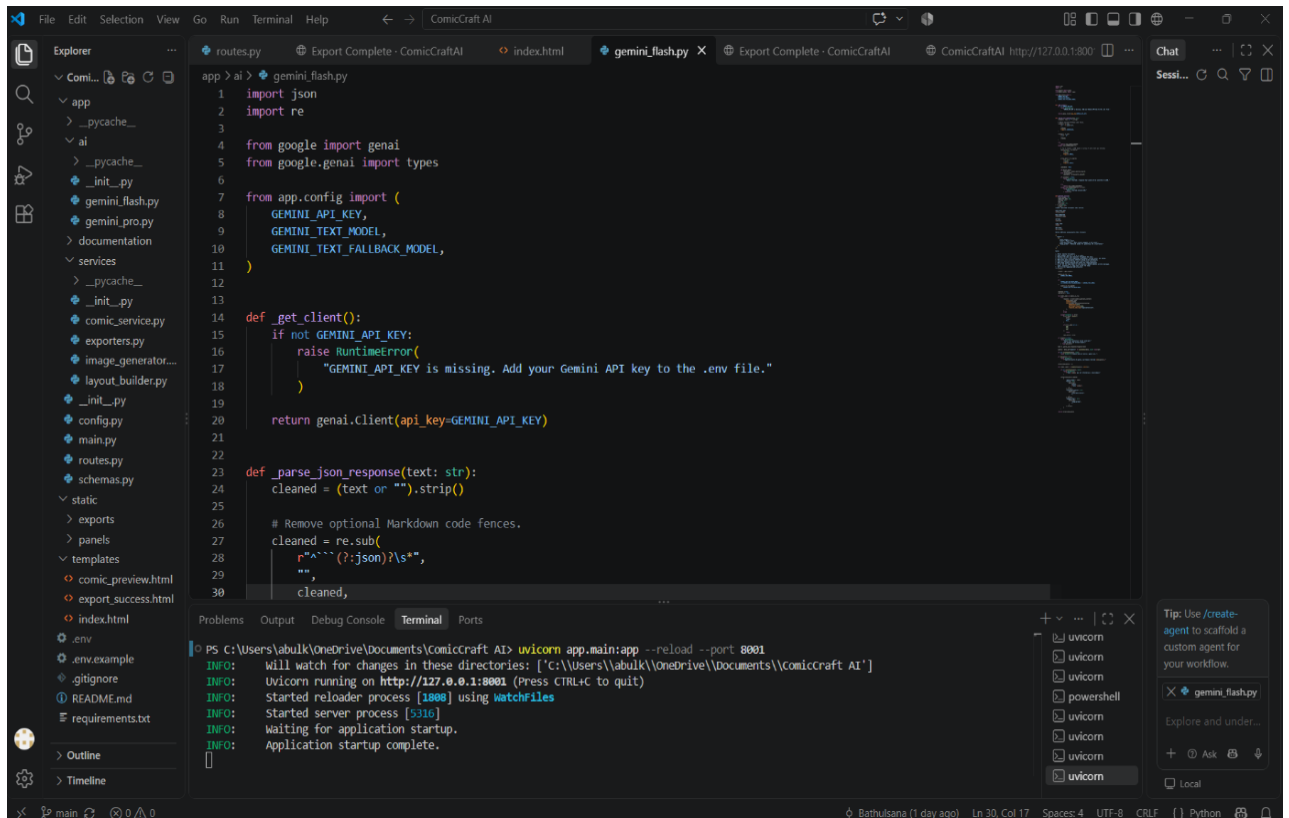
7. System Architecture

ComicCraftAI is organized into separate modules so that each major task has a clear responsibility.

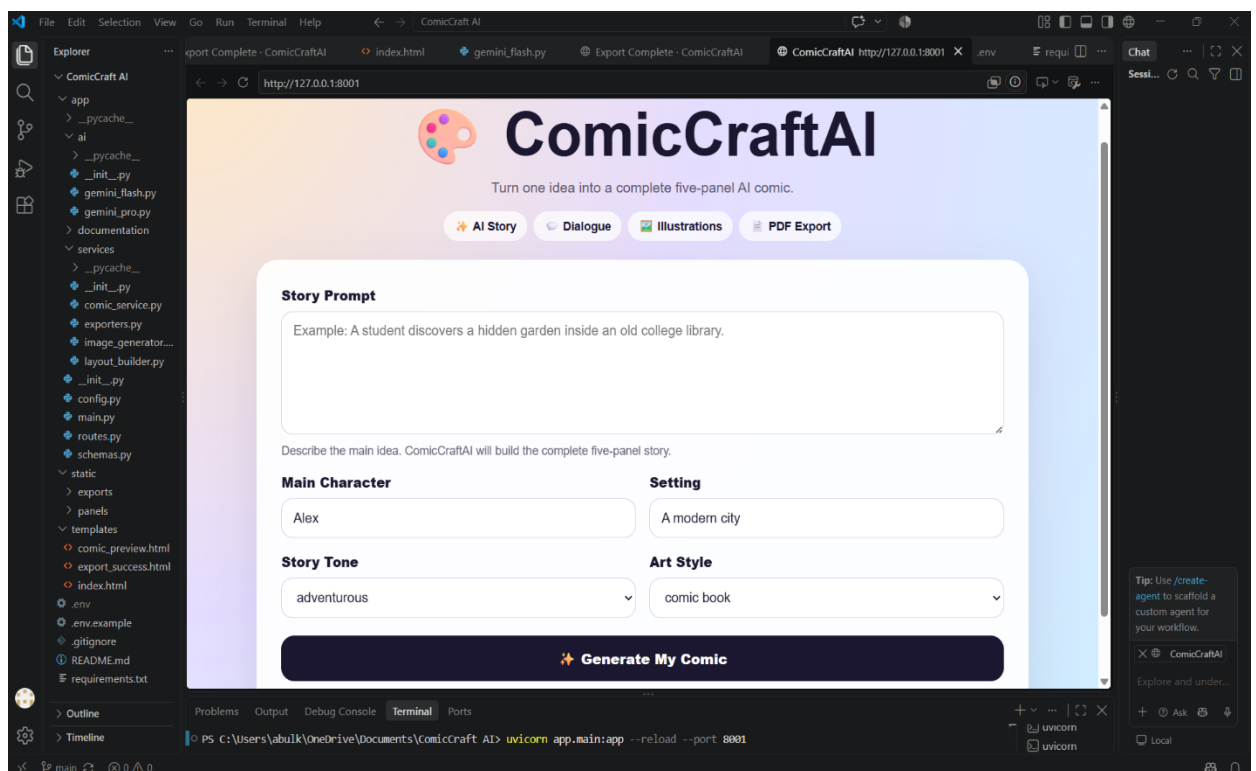
Overall Architecture



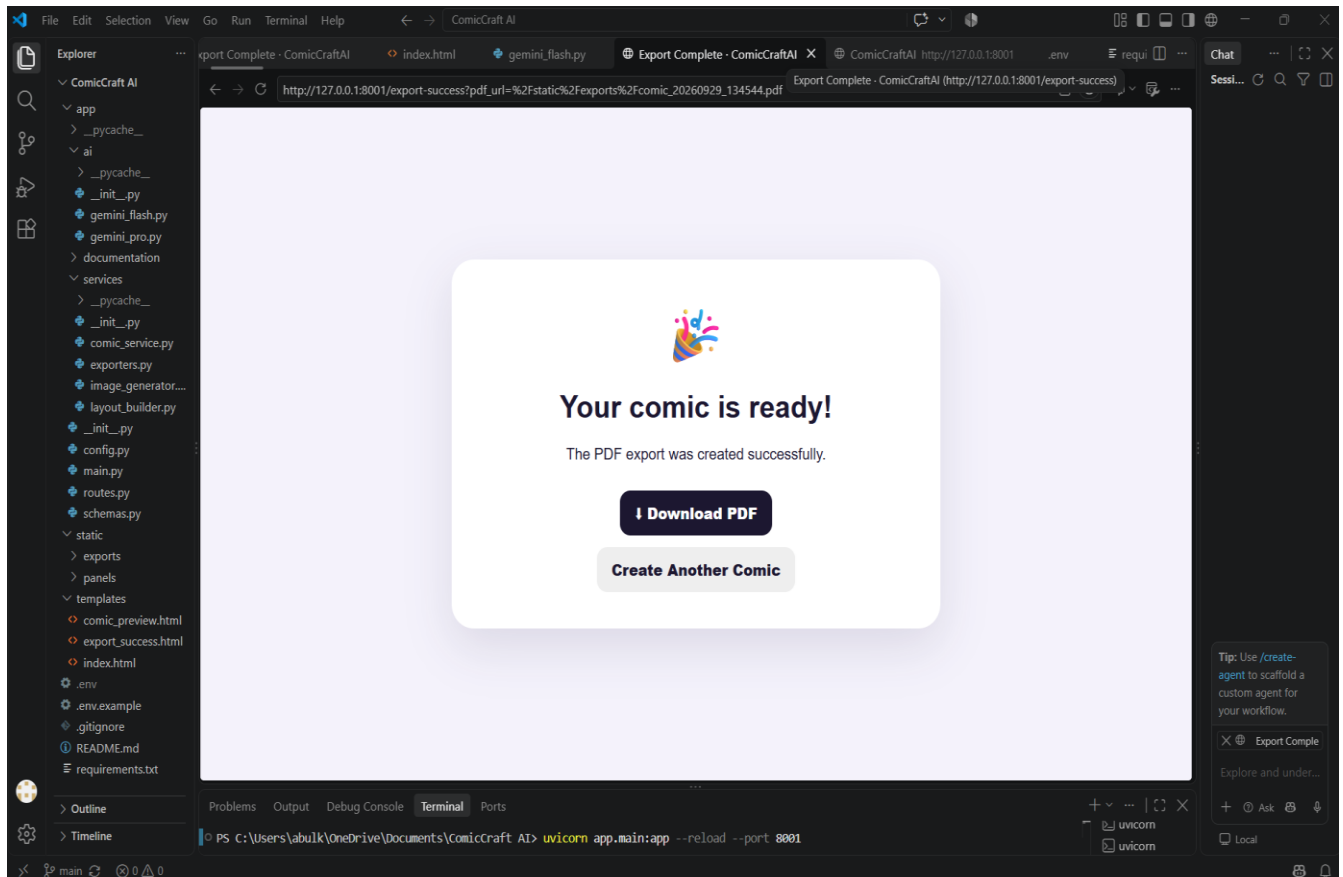
8. Project Structure



9. Web Interface



10.Final Outcome



The Completed Comic can also be exported as a PDF.

11.Major Project Modules

gemini_flash.py

Responsible for creating the initial five-panel story outline.

It takes the user's story idea and produces structured information for each panel, including the title, scene description and image prompt.

gemini_pro.py

Develops the outline into the actual comic narrative, including captions, narration and dialogue.

`image_generator.py`

Creates the visual artwork for each panel using the generated image prompts.

Our implementation uses hosted AI inference rather than requiring the user to install a large local image-generation model.

`comic_service.py`

Acts as the main workflow coordinator.

It connects the individual stages.

This is important because the modules do not work as isolated features; this service connects them into one complete comic-generation pipeline.

`layout_builder.py`

Combines the generated story information and image paths into a structured panel layout.

`exporters.py`

Creates the final PDF from the completed comic layout.

12. Working Demonstration — Sana Story

For our demonstration, we used the following input:

Story Prompt

“A girl is walking on the road and there she saw a old woman. She helped that old woman to cross the road. That old woman thanked her and gave her a bag full of apples. The Girl happily returned home.”

Input Details :-

Main Character: Sana

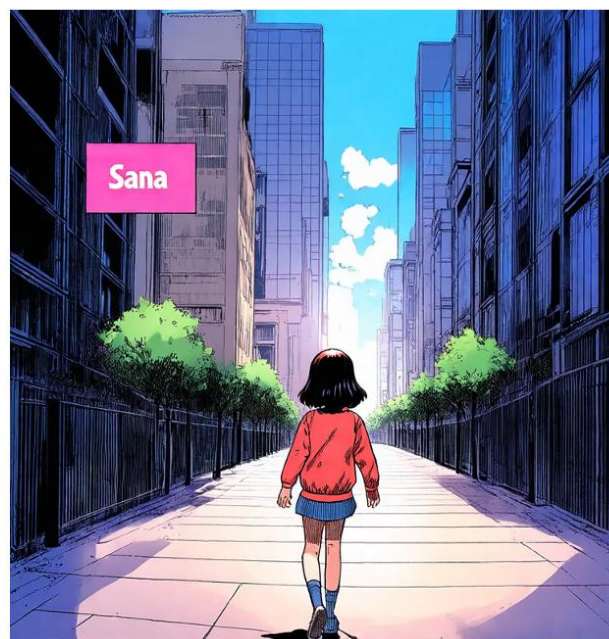
Setting: A Modern City

Tone: Heartwarming

Art Style: Comic book

From this single idea, ComicCraftAI generated a **five-panel storyline**, developed the narrative and dialogue, generated illustrations for the scenes, and combined the results into a comic that could be previewed and exported as a PDF

Panel 1: The City Walk



Sana, a young girl with a cheerful expression, is walking down a bustling modern city sidewalk filled with tall glass buildings and warm afternoon sunlight.

Caption: The City Walk

A bright afternoon in the bustling city brings a spring to Sana's step.

Panel 2: The Elderly Woman



Sana pauses near a busy crosswalk and notices an elderly woman holding a walking cane, looking hesitant to cross the bustling street.

Caption: The Elderly Woman

At the crosswalk, Sana notices someone who could use a little assistance.

Panel 3: A Helping Hand



Sana gently holds the elderly woman's arm, guiding her safely across the street while stopping traffic with a kind smile.

Caption: A Helping Hand

With practiced kindness, Sana steps in to guide her safely across.

Dialogue

Sana: Take your time, I'll walk with you.

Panel 4: A Grateful Gift



Standing safely on the opposite sidewalk, the smiling elderly woman hands Sana a heavy canvas bag filled with bright red apples as a token of gratitude.

Caption: A Grateful Gift

Safely across, the woman offers a sweet token of appreciation.

Dialogue

Elderly Woman: Thank you, dear. Please take these fresh apples.

Panel 5: Happy Homecoming



Sana skips down a residential city street toward her home during golden hour, holding the bag of apples tightly and beaming with joy.

Caption: Happy Homecoming

As the golden hour sets in, Sana heads home with a full heart and heavy bag.

13. Benefits

- Makes comic creation easier for beginners.
 - Reduces the time required for story planning and visual creation.
 - Combines text generation and image generation in one workflow.
 - Allows users to customize the character, setting, tone and art style.
 - Produces a structured five-panel comic rather than isolated AI-generated images.
 - Provides a downloadable PDF that can be saved or shared.
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14. Future Enhancements

Possible future improvements include:



Better Character Consistency

- A character-reference system could help maintain the same appearance across every panel.



Custom Panel Layouts

- Users could choose different comic layouts instead of having one fixed arrangement.



Automatic Speech Bubbles

- Dialogue could be placed directly into speech bubbles within the artwork.



Multiple Language Support

- Users could create comics in different languages.



Panel Editing

- Users could regenerate or edit an individual panel without generating the entire comic again.



More Export Options

- Additional formats such as image collections or presentation-ready files could be supported.

Online Deployment

- The application could be deployed online so users can access it without setting up the development environment locally.
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15. Conclusion

ComicCraftAI demonstrates how generative AI can be integrated into a complete creative application rather than being used only as a standalone chatbot or image generator.

The project takes a simple story idea and processes it through multiple stages—**story planning, narrative generation, image generation, layout construction and PDF export**—to produce a complete comic.

Through this project, we explored the practical integration of AI services with a Python web application and learned how different software components can work together as one workflow.

Our aim was to make the process **simple for the user while keeping the underlying workflow structured and modular.**

16. Team Details

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17. GitHub Repository

The source code and project files are maintained in our GitHub repository:

ComicCraftAI — GitHub Repository

View GitHub Repository :

<https://github.com/Bathulsana/ComicCraftAI>

The repository contains the application's source code, templates, configuration examples, requirements and project documentation.
